

# NGUYEN MAI HONG TRAM

## AI Researcher

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## Research Profile

My research has evolved from computer vision and medical visual question answering toward multimodal and video understanding, with recent work on temporally grounded action understanding in egocentric video. I am particularly interested in how vision-language models can move beyond static perception to reason about **actions, time, and embodied interaction**.

## Education

### B.Sc. in Data Science

University of Economics Ho Chi Minh City (UEH)

Sep 2022 – Mar 2026

GPA: 3.92/4.00

- **Rank 2** in Data Science; **Top 5%** of the School of Business Information Technology.
- Merit Scholarships for outstanding academic and extracurricular performance, 2023 and 2025.
- Selected coursework: Data Structures & Algorithms (**9.8/10**), Mathematics for Computer Science (**9.8/10**), and Artificial Intelligence (**10.0/10**).

## Selected Programs

### Applied AI Talent Program — VinGroup × VinUniversity

Intensive AI training, applied capstone, and industry research internship at VinRobotics

May 2026 – Aug 2026

### AIO2026 — AI VIETNAM

Research-oriented artificial intelligence training and community activities

Jun 2026 – Present

## Research & Industry Experience

### AI Research Intern

VinRobotics — AI Platform, VLA Team

Jul 2026 – Aug 2026

Hanoi, Vietnam

- Developed and evaluated multi-stage vision-language pipelines for **temporally grounded action captioning** on egocentric video.
- Designed controlled experiments on frame sampling, temporal grounding, prompting, and model adaptation; analyzed failure modes including directional ambiguity, dataset shift, and overfitting.
- Investigated recent VLM/VLA methods through literature review, implementation analysis, and large-scale experimentation using **SGLang**.

### AI Research Engineer Intern

TMA Solutions

Mar 2026 – May 2026

Ho Chi Minh City, Vietnam

- Investigated deep-learning approaches for bidirectional sign-language communication through literature review, algorithm analysis, implementation, and experimental evaluation.
- Researched continuous sign-language recognition and text-to-3D-avatar generation, achieving approximately **5% sequence prediction error** on the test set and **10%** in a real-time video-chat setting.
- Translated research prototypes into a browser inference pipeline using **PyTorch** → **ONNX Runtime Web**, reducing inference time by approximately **37%**.

### Research Collaborator

University of Economics Ho Chi Minh City

Mar 2025 – Jan 2026

Ho Chi Minh City, Vietnam

- Conducted research across **surgical VQA**, **Vietnamese scene-text recognition**, and **multi-objective optimization**.
- Contributed throughout the research lifecycle: problem formulation, literature review, data preparation, model development, experimental design, evaluation, and manuscript preparation.

Publications

GNN-SurgVQA: Object-Centric Graph Reasoning for Visual Question Answering in Laparoscopic Scene Understanding

International Conference on Data Analytics and Management (ICDAM), 2026 — *Best Paper* · [Code]  
Object-centric multimodal reasoning for surgical VQA; evaluated on **960K+ QA pairs**, achieving **85.9% accuracy** and **85.8% F1**.

Comprehensive Approach to Vietnamese Scene Text Recognition: Challenges, Models, and Framework Development

ICDAM 2025, Lecture Notes in Networks and Systems, vol. 1601, Springer, 2025 · [Paper]  
End-to-end Vietnamese scene-text detection and recognition; fine-tuning on approximately **25K samples** improved recognition performance by more than **2%**.

Selected Projects

Multi-Objective Optimization for Emergency Blood Transportation [Code]

Research Project · UEH Young Researchers Award — Prize B · Eureka 2025 Semifinalist

- Formulated emergency blood transportation as a multi-objective **EVRP with battery-swapping stations and time windows**, jointly optimizing logistics cost and hospital waiting time.
- Developed hybrid **NSGA-II/SPEA2** methods with Savings, 2-opt, and 3-opt heuristics.
- Validated on standard EVRP-TW benchmarks and a real Ho Chi Minh City case study involving **36 hospitals and 6 battery-swapping stations**.

GradeMind — AI-Assisted Mathematics Grading Platform [Demo]

Capstone Project · VinGroup × VinUniversity Applied AI Talent Program

- Built a teacher-in-the-loop AI system for grading Vietnamese secondary-school mathematics with evidence-backed feedback and class-level error analysis.
- Developed a multimodal grading and RAG pipeline with confidence-based routing, source-grounded evidence, PII safeguards, and golden-set evaluation.
- Stack: **FastAPI · PostgreSQL/pgvector · Celery · Redis · React/TypeScript · Docker**.

Technical Skills

|                    |                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Research           | Multimodal Learning, Vision-Language Models, Video Understanding, Computer Vision, Visual Question Answering, Model Evaluation |
| Machine Learning   | PyTorch, Hugging Face Transformers, PyTorch Geometric, scikit-learn                                                            |
| ML Systems         | SGLang, ONNX Runtime, FastAPI, Docker                                                                                          |
| Programming & Data | Python, SQL, PostgreSQL, pgvector                                                                                              |
| Language           | English — TOEIC Listening & Reading <b>930/990</b>                                                                             |